

Design and Implementation of a Student Attendance System Using Iris Biometric Recognition

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Abstract— Attendance taking is a standard practice in every educational system. The methods used to take class attendance are quite numerous but emphasis keeps shifting towards automating the process. The use of biometrics in taking class attendance is fast gaining ground and the traditional way of taking attendance is fast losing ground especially when the class is very large and time is of great essence. The iris was used as the biometric in this paper. After enrolling all attendees by storing their particulars along with their unique iris template, the designed system automatically took class attendance by capturing the eye image of each attendee, recognizing their iris, and searching for a match in the created database. The designed prototype is also web based. This paper proposes an alternative and accurate method of taking attendance that is both spoof-proof and relatively cheap to implement.

Keywords—Biometrics, iris, recognition, web application, attendance.

I. INTRODUCTION

All formal institutions of learning put a great premium on class attendance and it is so vital that students who do not meet the benchmark assigned to class attendance are not allowed to sit for examination. An automated class attendance system that uses iris biometric to take class attendance is the focus of this paper. The attendance of students in a class is tracked and automatically updated each time their iris gets scanned before entering the class. The use of the iris as a biometric is quite popular and favored by many researchers due to its near

permanence. The iris is also protected by the cornea and eyelid [1, 2].

The random patterns of the iris can be said to resemble a barcode. Its series of tangled connective tissue meshwork is never the same even in multi-birth situations [3, 4]. This uniqueness is the utilized characteristic when the iris is used as a biometric [5]. The use of iris as a biometric is made possible by high resolution images of the iris taken by powerful eye scanners.

The iris' uniqueness was utilized in this paper as a means of automating the process of taking class attendance with the aid of a database and supportive hardware. The designed system allows students to be registered or enrolled and verified when attendance is taken. The use of iris as a biometric in taking class attendance eliminates the issues of forgery and impersonation.

During enrolment, the designed graphical user interface was used to obtain the student particulars and get the iris image captured by the scanner. Each unique iris template, along with the corresponding particulars was stored in a database. Attendance was taken automatically (database updated) when class attendees were verified after getting their iris captured by the iris scanner.

II. RELATED WORKS

A. How Iris Recognition Works and Stages of the Iris Recognition System

The colour tissue of the iris surrounding the pupil has more than 200 points that are unique to every individual. The points include rings, furrows and freckles [5].

During enrolment, the eye is first photographed. An algorithm then creates a template of the eye's iris pattern and stores it in a database. When identifying, the eye image is taken and the iris template created is compared with the templates stored in the database for a match. If a match is found, the individual is identified. If there is no match, the individual is not identified. The processes from capturing the user's eye image to identification are elucidated below:

- *Image Acquisition:* A very high resolution image of the eye is acquired with the help of an iris scanner. The resolution of the iris image is such that it eases the subsequent steps. The contrast and brightness of the acquired eye image is taken into consideration. The captured eye image is usually well framed during segmentation; the edges of the iris and pupil are mapped out using Daugman's Integro-differential equation. The process of mapping out of the edges of the iris and separating it from other parts of the eye is called Iris Localization. The next stage in the segmentation process is called Normalization and it is the process of standardizing the size of the localized iris to cater for variations in iris sizes caused by pupil dilation and constriction [6].
- *Iris segmentation:* The iris is located and separated from the white sclera and pupil.

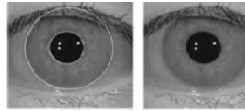


Fig. 1. The pupil and iris respectively.

B. Iris Feature Extraction

The unique features of the iris are extracted using sophisticated image processing techniques to create a digitized template of the iris that can be stored in a database [7].

C. Iris Code Matching

Comparisons are made between an unknown iris image and reference images stored in the database for any match based on the pre-set threshold.

III. LITERATURE REVIEW

Wireless Attendance Management System Based on Iris Recognition written by [8] noted the fact iris recognition was once a science fiction theme. It also reaffirmed the uniqueness

of the iris of every individual. The goal of the paper was to incorporate biometrics into a wireless attendance system. The design had the potential to improve work efficiency and reduce the rate of absenteeism but it needs a very large network bandwidth and a high startup cost to implement it.

Student Management through Iris Recognition System proposed by [3] with a firm declaration that iris was the best biometric for authentication and verification purposes. The paper suggested the use of neural networks for pattern matching and image processing. The aim of the paper was to create a web portal through which student attendance can be managed by using the iris as biometric for identification purposes. It is a very efficient way of tracking students' attendance but requires highly skilled personnel and huge initial capital outlay to set up.

Secured Biometric Verification Based on Iris Recognition proposed by Adam Czajkaa,b and Krzysztof Piecha from the Institute of Control and Computation Engineering, Warsaw University of Technology, Warsaw, Poland. They proposed a secure biometric verification station that had a camera, processing unit, iris recognition and processing software and a LCD display. It used a specially designed protocol to control access to the station and communicate with the external world. They employed an application of Zak-Gabor based iris coding to do it. The main aim of the paper was to propose a secure verification station using the well-established Zak-Gabor based iris coding. The accuracy of the system was very high and fully automated but cost and scalability was not favourable [9].

Iris Based Attendance System proposed by a group of Professors Prashik S. Bhagat, Prof. D. S. Shilwant, Prof. S. P. Kharde, Praful S. Bhagat, Abhijit S. Andure, Prof. Amol A. Shirsath. They proposed an attendance system that uses the iris as a biometric for enrolment and identification [6].

Iris Recognition System proposed by Neha Kak, Rishi Gupta Computer Science Engineering Lingayas Institute of Technology and Management Faridabad, Haryana, INDIA and Sanchit Mahajan. The main objective of their work was to identify a person in real time with the Canny Edge detector algorithm. They proposed that this mode of iris recognition could be applied to automated passports, aviation security, database access, computer login and controlling access to restricted areas.

IV. METHODOLOGY

An Iris Scanner was used get the image of the iris. The graphical user interface (webpage) was made using HTML, JavaScript, and CSS. The graphical interface enabled effective and easy communication between users and the designed system. The database was created using MySQL language.

Students are identified when a match is found in the database for the iris image of each student just acquired. The designed system returns an integer value of (1) if there is a match and an integer value of (0) if there is no match. These results are stored in the database along with the time, date, month, and year the attendance was taken.

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The database was designed using MySQL and can store at the most, ten thousand (10,000) entries. The database is shown in Figure 2 and 3. It consists of the attendance and profile database.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	ID	int(11)			No	None	AUTO_INCREMENT		Change Drop Primary Unique Index More
2	Username	varchar(20)	latin1_swedish_ci		No	None			Change Drop Primary Unique Index More
3	Password	varchar(20)	latin1_swedish_ci		No	None			Change Drop Primary Unique Index More
4	AccountType	int(11)			No	None			Change Drop Primary Unique Index More
5	status	int(11)			No	None			Change Drop Primary Unique Index More
6	date	datetime			No	None			Change Drop Primary Unique Index More

Fig. 2. Attendance database.

ID	loginID	surname	lastname	othernames	phone	email	regnumber	gender
6	0	Osahie	Samuel		08055989916	Osahiesamuel@gmail.com	12CA13715	60216d78163a6a3c210404
5	0	Ordu	Peace		08110707950	peaceordu@yahoo.com	12CA13710	4761f8cc37008453925c4b
2	0	Madu-Oruohar	Makuchukwu	Chukwizom	08110707950	makumadu@yahoo.com	12CA13365	321e475ab70ab6d413a6d
7	0	Jumbo	prince	Chambaran	08056914100	princejumbo3@yahoo.com	12CA13363	8623977a6030a75a107336d
3	0	Ekpe	Chinonso	Okwada	7067344485	ekpechinonso@yahoo.com	12CA13371	4865d78163a6a3c210404

Fig. 3. Profile database.

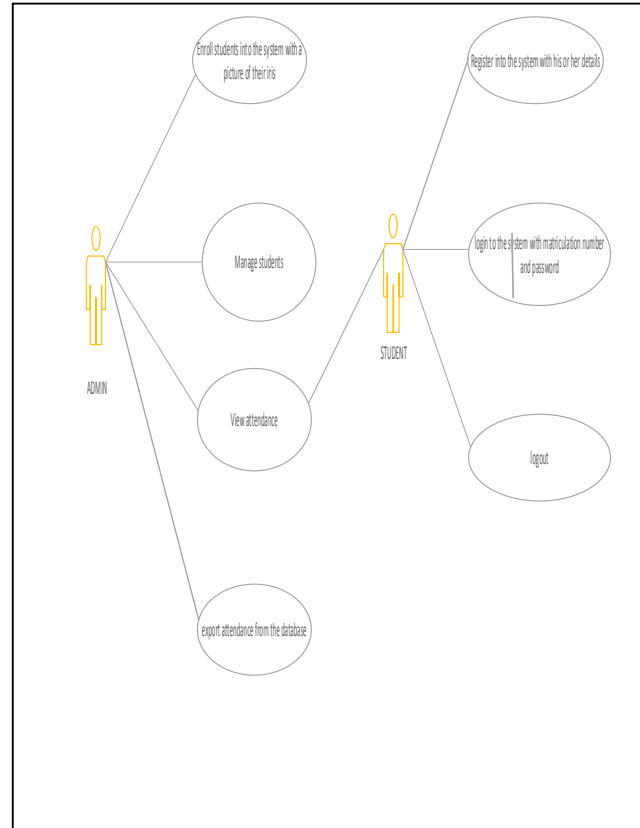


Fig. 4. Use case diagram.

A. Identification

All students are first enrolled by taking their particulars and storing it along with their unique digital iris template taken by the iris scanner in the database.

When identifying students or taking attendance, the iris scanner is launched from the web portal. All students attending a class place one of their eyes close to the scanner.

The scanner takes high resolution images of their irises and creates a digital template of the iris. The designed system searches for a match in the database for each iris template and marks any match as present.

The administrator can see the overall attendance and each student can view their respective attendance.

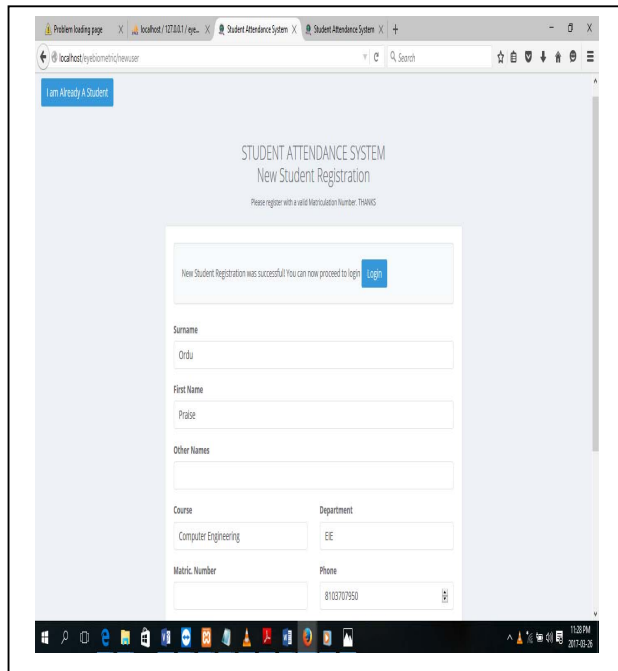


Fig. 5. Registration page

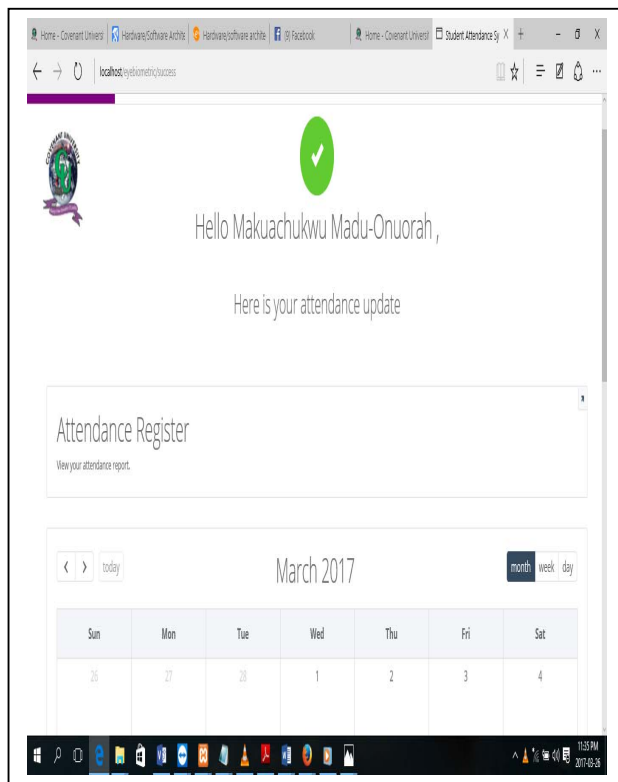


Fig. 6. Attendance window for students

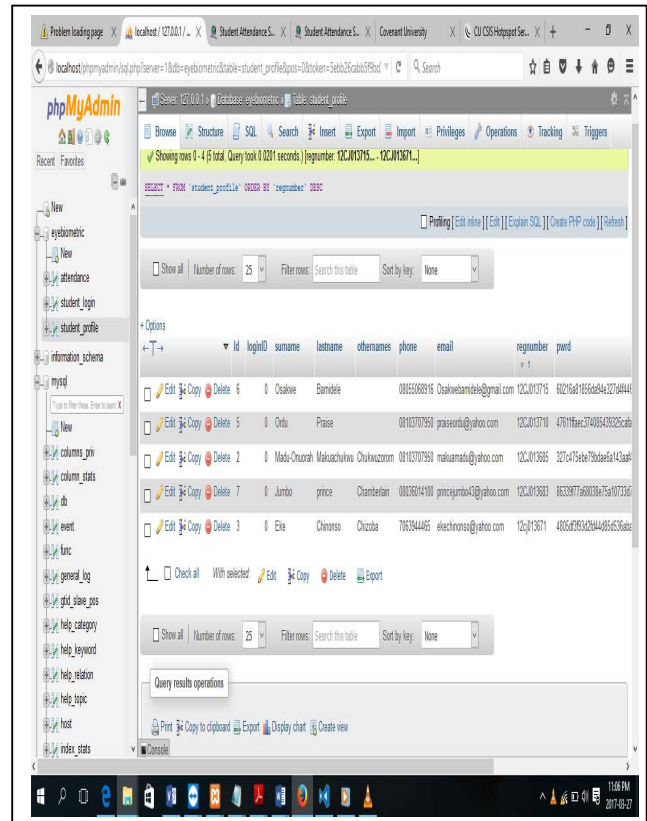


Fig. 7. Attendance window for administrators

V. RESULT AND DISCUSSION

The designed student attendance system's performance was measured using the following is measured in terms of the following [11].

A. False Match Rate (FMR):

This is the percentage of times the student attendance system produces a false accepts, which occurs when an individual is incorrectly matched to another user. From tests carried out with the system, there was no false match, as access wasn't granted to wrong users. (FMR = 0%)

B. False Positive Identification Rate (FPIR):

This occurs when the student attendance system finds a hit for Query iris pattern that has not been enrolled into the system.

$$FPIR = 1 - (1 - FMR)^N$$

Where N= number of students enrolled into the system = 5

FMR = 0

$$FPIR = 1 - (1 - 0)^5 = 1 - (1)^5 = 1 - 1 = 0$$

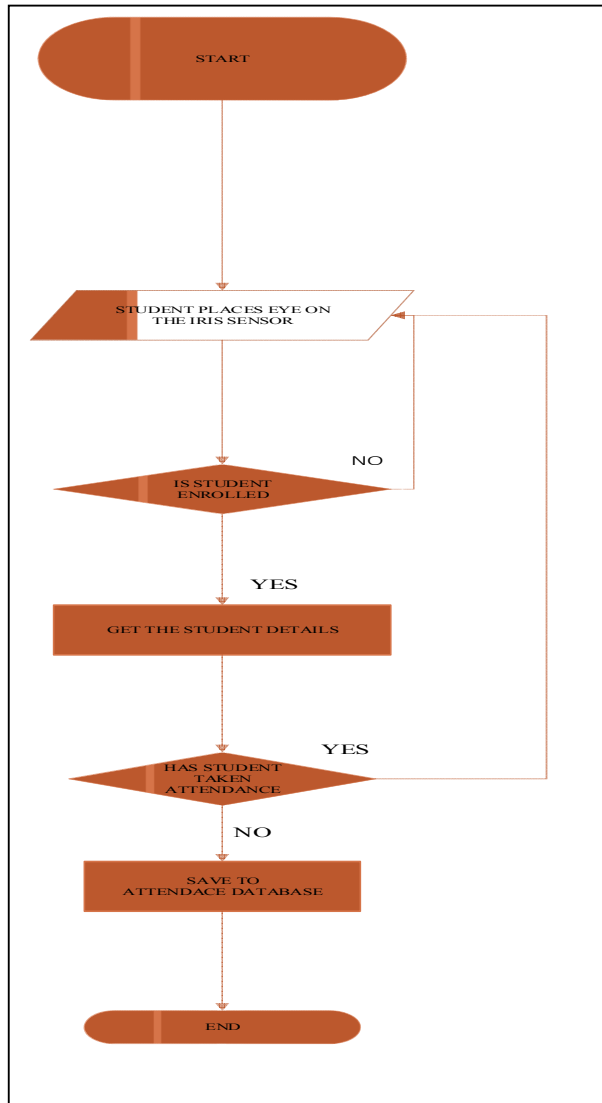


Fig. 8. Algorithm for designed iris student attendance system.

C. False Negative Identification Rate (FNIR):

This occurs when the student attendance system finds no hit or a wrong hit for a query iris pattern enrolled into the system.

$$FNIR = 1 - (1 - FNMR)^N$$

N= number of students enrolled into the system = 5

FNMR = 0

$$FNIR = 1 - (1 - 0)^5 = 1 - (1)^5 = 1 - 1 = 0.$$

VI. CONCLUSION

The main aim of this paper is to propose a cheap and feasible automated attendance system using the iris as the biometric for enrolment and identification. The attendance system proposed by this paper is web based. The designed prototype showed a lot of promise and presents an alternate and more accurate method of taking attendance that is quite spoof proof and relatively cheap to implement.

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